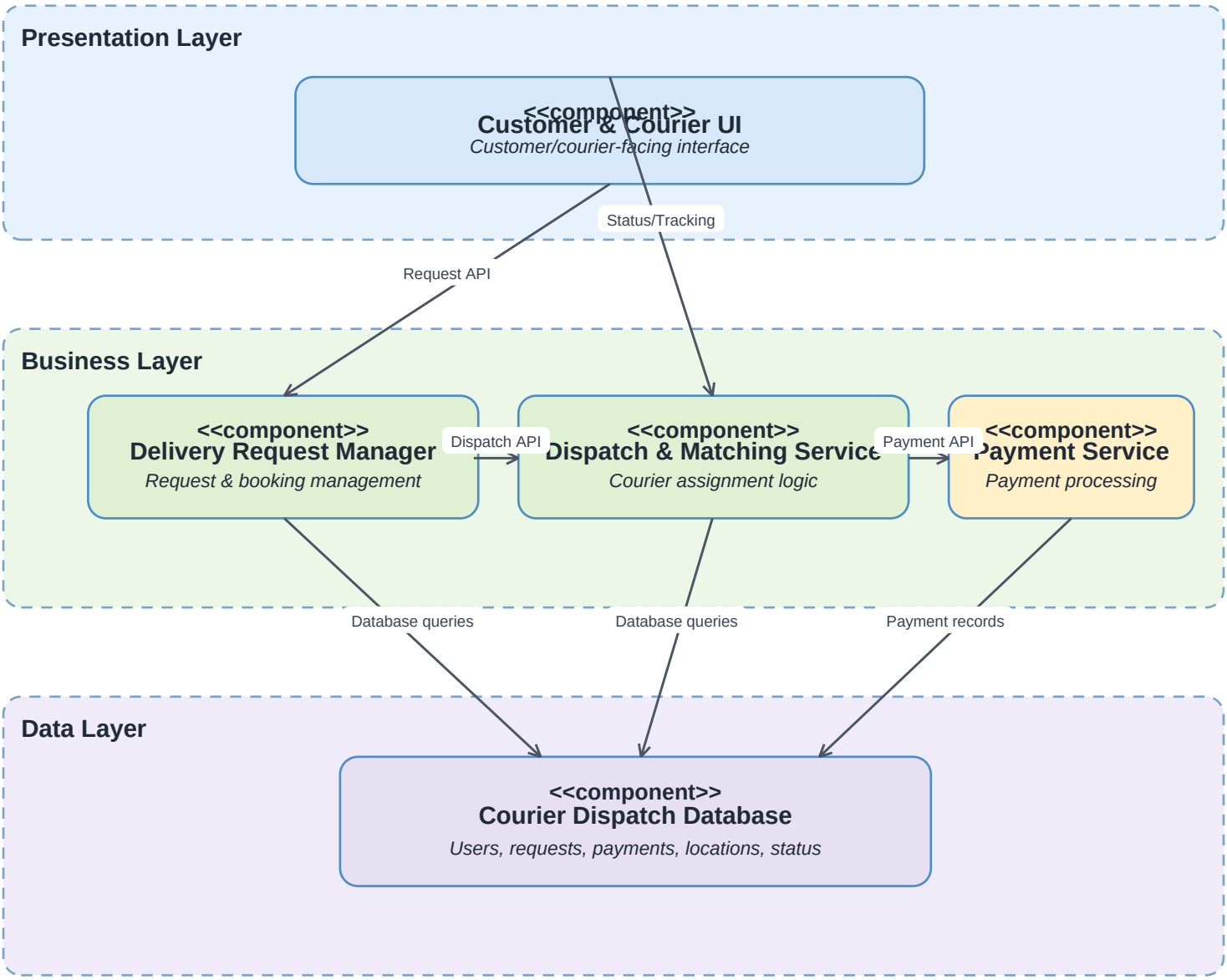
LAB 3 - COMPONENT DIAGRAM

Hyperlocal Courier Dispatch & Tracking Engine

Arya MG | SRN: PES2UG24CS087 | Selected Architecture: Layered



UML notation: components represent modular system responsibilities; labelled connectors show service/data flow.

LAB 3 - ARCHITECTURAL JUSTIFICATION

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Architecture Selection: We chose Layered Architecture for the Hyperlocal Courier Dispatch & Tracking Engine.

The system benefits from a clear separation between the customer/courier interface, delivery and dispatch business logic, and persistent data. The layered structure keeps the core workflow straightforward while allowing individual business components such as matching, payment and tracking to evolve independently.

Architectural Choice - Two Scenario-Related Reasons

1. Clear separation of responsibilities. The UI handles customer and courier interaction; the Delivery Request Manager, Dispatch & Matching Service and Payment Service handle business functions; and the Dispatch Database stores persistent delivery information. This makes the system easier to understand, test and maintain.

2. Simpler integration for the core workflow.

Creating a delivery request, calculating ETA, assigning a nearby courier, processing payment and updating delivery status require predictable interactions. A layered design provides direct, understandable dependencies without the deployment and operational complexity of a distributed microservices architecture.

Security Advantage

Security responsibilities can be concentrated in the business layer. The Delivery Request Manager can validate requests and authorization before allowing payment or database operations. Keeping database access behind business components also reduces direct exposure of stored delivery data.

Performance Benefit

The layered design avoids unnecessary network overhead for a lab-sized deployment. The Dispatch & Matching Service can optimize courier selection and ETA calculations without changing the customer-facing interface.

Component Identification

Layered components are identified according to their role in the delivery system.

Layer	Component	Purpose
Presentation	Customer & Courier UI	Customer-facing interaction for delivery requests, trac
Business	Delivery Request Manager	Creates and manages delivery requests and booking infor
Business	Dispatch & Matching Service	Finds suitable couriers and coordinates assignment and
Business	Payment Service	Handles payment processing and payment-related requests
Data	Courier Dispatch Database	Stores users, delivery requests, payments, locations an

Interfaces shown: Request API, Dispatch API, Payment API, Status/Tracking and Database Queries.